CS3570 Introduction to Multimedia Technology

Homework #3

Due: 11:59 pm, 2026/05/11

1. Bézier curve (50%)

You are given 1 raster image (bg.png) and 1 path object (Bézier curve, extracted as **N points** in points.txt); your objective is to draw the points and the Bézier curves.



bg.png

a. (30%) Plot two sets of **cubic Bézier curves** with different detail levels, along with their control points, onto the provided image bg.png. Each set contains (N−1)/3 cubic Bézier curves connected end-to-end, outlining the subject in the image.
   - For low detail curve: *t = {0, 0.25, 0.5, 0.75, 1.0}*
   - For high detail curve: *t = {0, 0.001, 0.002, …, 1.0}*

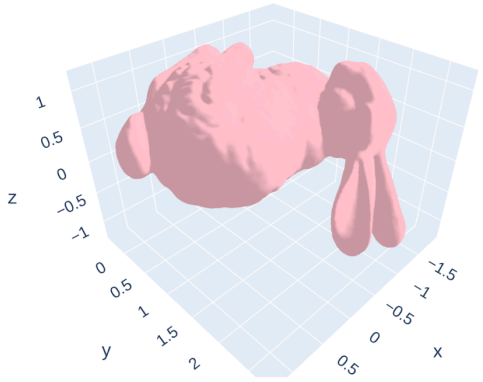
   Color the low-detail curve cyan and the high-detail curve magenta. Save the resulting image as **1a.png**.

b. (10%) Scale up the image and the curves by 4x, then plot and save the results as **1b.png**.
   I. bg.png is scaled using Nearest-neighbor interpolation.
   II. First, scale the N points, then draw the curve (using the settings of the high-detail curve).

c. Report
   I. Explain how you implemented the Bézier curve calculations. (3%)
   II. Compare the high and low-detail curves. Discuss your observations regarding their differences. (3%)
   III. Compare the differences between upscaling the raster background image and the vector curves. (4%)

2. 3D Models (50%)

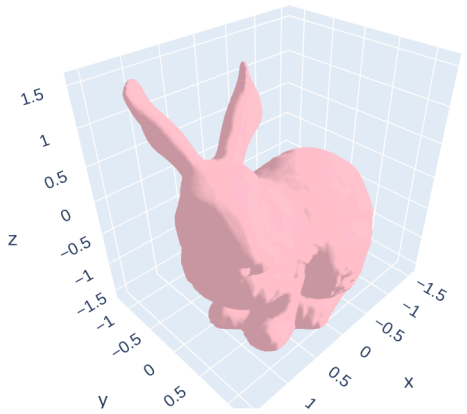
In this section, you will implement geometric transformations and experiment with different illumination settings on the 3D model bunny.obj. Here, we use Plotly, a Python graphing library, to render 3D graphics.



bunny.obj

a. Implement translation and rotation transformations to position and orient the bunny such that it satisfies **both** of the following conditions:
   I. The center of the bunny is placed at the origin `(0, 0, 0)`. The center is defined as `[(max(x) + min(x)) / 2, (max(y) + min(y)) / 2, (max(z) + min(z)) / 2]` where x, y, z are the vertices of the object. (10%)
   II. The bunny is rotated so that it is facing directly toward the screen (see the example below). (20%)

   Save the plot of the transformed 3D figure as **2a.png**.



b. (5%) Based on (a), render the bunny under different illumination settings by varying the ambient strength k_a, diffuse strength k_d, and specular strength k_s in the following settings.:
   I. (k_a , k_d , k_s) = (0.1, 0.5, 0.05) and (k_a , k_d , k_s) = (0.8, 0.5, 0.05)
   II. (k_a , k_d , k_s) = (0.4, 0.1, 0.05) and (k_a , k_d , k_s) = (0.4, 0.9, 0.05)
   III. (k_a , k_d , k_s) = (0.5, 0.8, 0.1) and (k_a , k_d , k_s) = (0.5, 0.8, 0.8)

   Plot the result for each condition with **2 subplots** and discuss the difference in each pair of settings. Results are saved as **2b_1.png, 2b_2.png, 2b_3.png,** respectively.
   You can refer to the following link for more information: **Plotly Lighting Reference**

c. Report
    I. Describe how you designed and implemented the transformations in 2(a). Does the order of the transformations you apply matter? Explain why. (5%)
    II. Discuss the visual differences observed in each pair of lighting settings in 2(b). Explain the effects of ambient, diffuse, and specular lighting based on your observations. (10%)

Reminder

- You are not allowed to use any function that directly generates the result for each step, except for those provided by us.
- Your code must display and output your results to enable us to verify its correctness.
- Please follow the instructions in the Jupyter Notebook and complete the parts marked as “TODO”.
- Please include your report in the Jupyter Notebook using markdown.
- Please compress your code, input images, and result images in a zip file named HW3_{StudentID}.zip and upload it to eeclass.
- Homework should be submitted before the announced due time. Scores of late submissions will be reduced by 20% per day.
- If you encounter any problems or have questions, please post them on eeclass.
- Please follow the file structure below:

```
HW3_{StudentID}
├── bg.png
├── bunny.obj
├── HW3.ipynb
├── points.txt
└── output
    ├── 1a.png
    ├── 1b.png
    ├── 2a.png
    ├── 2b_1.png
    ├── 2b_2.png
    └── 2b_3.png
```